

LSD: S-matrix projects out 1-particle asympt. states from Time-ordered prod. of fields

$2 \rightarrow n$ Scattering

$$|i\rangle = \sqrt{2\omega_1} \sqrt{2\omega_2} a_1^\dagger(-\omega) a_2^\dagger(-\omega) |\Omega\rangle$$

$|\Omega\rangle$: Ground state with NO particles

$$* |f\rangle = \sqrt{2\omega_3} \sqrt{2\omega_4} \dots \sqrt{2\omega_n} a_{p_3}^\dagger(\omega) \dots a_{p_n}^\dagger(\omega) |\Omega\rangle$$

* $|f\rangle = |i\rangle$ (this occurs if no scattering happens)
 $\rightarrow$ and so $|f\rangle$ does not contribute

$$\langle f | S | i \rangle = 2^{n/2} \sqrt{\omega_1 \dots \omega_n} \langle \Omega | a_{p_3}(\omega) \dots a_{p_n}(\omega) a_1^\dagger(-\omega) a_2^\dagger(-\omega) | \Omega \rangle$$

$$\phi(x) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} (a_p(t) e^{-ipx} + a_p^\dagger(t) e^{ipx})$$

For a time indep. Ham. H

$$a_p(t) = e^{iH(t-t_0)} a_p(t_0) e^{-iH(t-t_0)}$$

$$\langle \Omega | \phi(x, t = \pm\infty) | P \rangle = C e^{iKx}$$

$$* i \int d^4x e^{iPx} (\Box + m^2) \phi(x) = \tilde{P} \{ i \phi(x) \}$$

$$i \int d^4x e^{iPx} (\partial_t^2 - \vec{\nabla}^2 + m^2) \phi(x) \\ = i \int d^4x e^{iPx} (\partial_t^2 + |P|^2 + m^2) \phi(x) = i \int d^4x e^{iPx} (\partial_t^2 + \omega_p^2) \phi(x)$$

$$* \partial_t (e^{iPx} (i\partial_t + \omega_p) \phi(x)) = i e^{iPx} (\partial_t^2 + \omega_p^2) \phi(x)$$

$$\therefore i \int d^4x e^{iPx} (\Box + m^2) \phi(x) = \int dt \partial_t [e^{i\omega_p t} \int d^3x e^{i\vec{P}\vec{x}} (i\partial_t + \omega_p) \phi(x)]$$

For $\phi(x)$ above

$$\int d^3x e^{-i\vec{p}\cdot\vec{x}} (i\partial_t + \omega_p) \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2\omega_k}} \left(a_k(t) e^{-i\vec{k}\cdot\vec{x}} + a_k^\dagger(t) e^{i\vec{k}\cdot\vec{x}} \right)$$

$$= \int \frac{d^3k}{(2\pi)^3} \frac{d^3x}{\sqrt{2\omega_k}} \left(a_k \omega_k e^{-i\vec{k}\cdot\vec{x}} - a_k^\dagger \omega_k e^{i\vec{k}\cdot\vec{x}} \right) + " \omega_p "$$

$$= \int \frac{d^3k}{(2\pi)^3} \frac{d^3x}{\sqrt{2\omega_k}} \left[(\omega_p + \omega_k) a_k(t) e^{-i\vec{k}\cdot\vec{x}} + (\omega_p - \omega_k) a_k^\dagger(t) e^{i\vec{k}\cdot\vec{x}} \right]$$

* This all assumes $\partial_t a(t) = 0$ (true at $t = \pm \infty$)

$$\int d^3x \rightarrow \delta^3(\vec{p} - \vec{k}) + \delta^3(\vec{p} + \vec{k})$$

leading to $\omega_p = \omega_k$

so second term vanishes and we have

$$2\omega_p a_p(t) e^{-i\omega_p t} = \int d^3x e^{-i\vec{p}\cdot\vec{x}} (i\partial_t + \omega_p) \phi(x)$$

$$i \int d^4x e^{i\vec{p}\cdot\vec{x}} (\Box + m^2) \phi(x) = \int dt \partial_t \left\{ e^{i\omega_p t} \sqrt{2\omega_p} a_p(t) e^{-i\omega_p t} \right\}$$

$$i \int d^4x e^{i\vec{p}\cdot\vec{x}} (\Box + m^2) \phi(x) = \sqrt{2\omega_p} [a_p(\infty) - a_p(-\infty)]$$

* Going back to $\langle f | S | i \rangle = \sqrt{2\omega_1 \dots \omega_n} \langle \Omega | a_3(\infty) \dots a_1 a_2(\infty) \dots$
Time ordering allows us to write this as

$$\langle f | S | i \rangle = \sqrt{2\omega_1 \dots \omega_p} \langle \Omega | T \{ (a_3(\infty) - a_3(-\infty)) \dots (a_p^\dagger(-\infty) - a_p(\infty)) \} | \Omega \rangle$$

Since T-ordering will move $a_3(-\infty)$ so that $a_3(-\infty) | \Omega \rangle = 0$ etc....

$$\langle p_1 \dots p_n | S | p_1 p_2 \rangle = \prod_{i=1}^n \left(i \int d^4x_i e^{-i p_i \cdot x_i} (\Box_i + m_i^2) \right) \prod_{j=3}^n \left(i \int d^4x_j e^{i p_j \cdot x_j} (\Box_j + m_j^2) \right) \times \langle \Omega | T \{ \phi(x_1) \dots \phi(x_n) \} | \Omega \rangle$$

LSZ-reduction formula

outgoing: $e^{-i p x}$
incoming: $e^{i p x}$

LSZ for Operators:

$$\langle p_1 \dots p_n | S | p_1 p_2 \rangle = \left\{ i \int d^4 x_1 e^{-i p_1 x_1} (\Box_1 + m^2) \right\} \dots \left(i \int d^4 x_n e^{i p_n x_n} (\Box_n + m^2) \right) \times \langle \Omega | T \{ \phi_1(x_1) \phi_2(x_2) \phi_3(x_3) \dots \phi_n(x_n) | \Omega \rangle$$

ϕ : operator which creates 1-part. states i.e.
 $\langle p | \phi(x) | \Omega \rangle = \int e^{i p x}$

• $\phi(x)$ can be defined by their matrix elements in a basis ψ_n

$$C_{nm} = \langle \psi_n | \phi | \psi_m \rangle$$

$$\begin{cases} |\psi\rangle = \sum_i^n \alpha_i |e_i\rangle \\ \langle e_j | A | \psi \rangle = \sum_i \alpha_i \langle e_j | A | e_i \rangle \end{cases}$$

$$\phi = \sum_{n,m} \int d^4 q_1 \dots d^4 q_n d^4 p_1 \dots d^4 p_m a_{q_1}^\dagger \dots a_{q_n}^\dagger a_{p_1} \dots a_{p_m} C_{nm}(q_1, \dots, p_m)$$

* Assume the interacting theory has a unique ground state $|\Omega\rangle$ s.t.
 $H|\Omega\rangle = 0$
 $P^\mu|\Omega\rangle = 0$

(i) First state is a particle $a^\dagger|\Omega\rangle$ with
 $E = \omega = (\vec{k}^2 + m^2)^{1/2}$

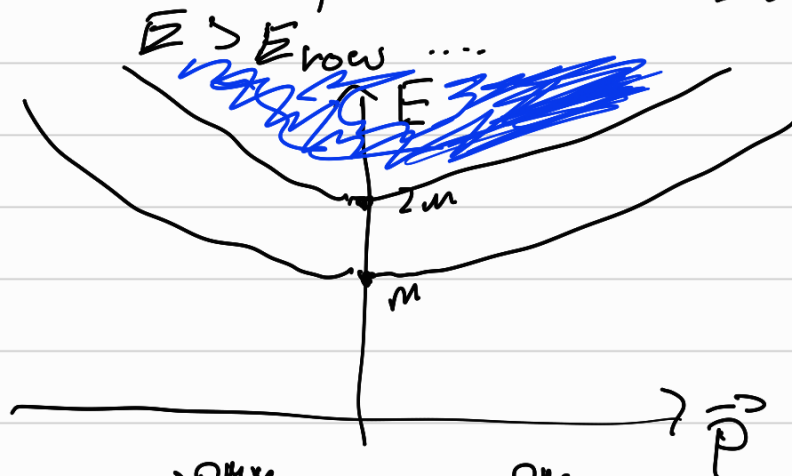
(ii) S_{21} (1 to 2): $a^\dagger a^\dagger |\Omega\rangle$: two particles

state

- Lowest possible state is zero momenta and s_0

$$E_{\text{low}} = m + m = 2m \quad (\text{Rest Masses})$$

- $\exists$ a frame of reference s.t. in the P.O.V. of the two particles $E_{\text{low}} = 2m$ but Lab frame



$$\psi(x) = e^{-ip_m x_m} \psi(0) e^{ip_m x_m}$$

$$\left[\begin{array}{l} \text{In 1D QM} \quad p^0 = H \quad \text{so} \\ \psi(x) = e^{-iHt} \psi(0) e^{iHt} \end{array} \right]$$

$$\langle \Omega | \psi(x) | \Omega \rangle = \langle \Omega | \psi(0) | \Omega \rangle$$

Lorentz-inv.
number

Since $p_m |0\rangle = 0$

$$U(\Lambda)^{-1} \psi(x) U(\Lambda) = \psi(\Lambda^{-1} x)$$

$x=0$:

$$U(\Lambda)^{-1} \psi(0) U(\Lambda) = \psi(0)$$

$$\psi(0) U(\Lambda) = U(\Lambda) \psi(0)$$

*Want $\langle \Omega | \psi(0) | \Omega \rangle$ to be zero as $a^\dagger(\pm\infty)$ acting on $|\Omega\rangle$ to create a 1PS.

- Let $v = \langle \Omega | \psi(0) | \Omega \rangle \neq 0$, then we can shift our field $\psi(x)$ by v

$$\psi(x) \mapsto \tilde{\psi}(x) = \psi(x) + v$$

Replace $\psi(x)$ in the Lagrangian

$\hat{\psi}$ now obeys $\langle \Omega | \hat{\psi} | \Omega \rangle = 0$

* Now project the momenta from $\hat{\psi}(x) | \Omega \rangle$

$$\begin{aligned} \langle p | \hat{\psi}(x) | \Omega \rangle &= \langle p | e^{-i \hat{p}^\mu \hat{x}_\mu} \hat{\psi}(0) e^{i \hat{p}^\mu \hat{x}_\mu} | \Omega \rangle \\ &= e^{-i p^\mu \hat{x}_\mu} \langle p | \hat{\psi}(0) | \Omega \rangle \end{aligned}$$

$$\langle p | \hat{\psi}(x) | \Omega \rangle = e^{-i p x} \underbrace{\langle p | \hat{\psi}(0) | \Omega \rangle}_{\text{Lorentz inv. number}}$$

which has to be a p^μ of $p^2 = p^\mu p_\mu$

Since this is the only Lorentz inv. comb. which is a constant $p^\mu p_\mu = -m^2$

$$\langle p | \hat{\psi}(0) | \Omega \rangle (m^2)$$

* We would like $\langle p | \hat{\psi}(0) | \Omega \rangle$ to be 1 as this would correspond to the creation of a 1PS.

$$\langle p | \hat{\psi}(0) | \Omega \rangle = \langle p | \psi(0) | \Omega \rangle + V \langle p | \Omega \rangle$$

Rescale $\hat{\psi}(x)$ by a multiplicative constant s.t.

$$\langle p | \hat{\psi}(0) | \Omega \rangle = 1$$

$$\hat{\psi}(x) = \frac{\psi(x)}{\langle p | \psi(0) | \Omega \rangle} \quad (\text{Renormalize})$$

* Consider $\langle p, n | \hat{\psi}(x) | \Omega \rangle$

$|p, n\rangle$: multiparticle state w total three momenta $\vec{p}$

n : all other labels

$$\langle p, n | \hat{\psi}(x) | \Omega \rangle = e^{-i p x} \langle p, n | \hat{\psi}(0) | \Omega \rangle$$

$$\langle p, n | \psi(x) | \Omega \rangle = e^{-i p x} A_n(\vec{p})$$

$A_n(\vec{p})$: f^n of L.T. products of relative & total 4-momenta

$$* \quad 1 \rightarrow n \quad n \geq 2$$

So we would want $\langle p, n | \psi(x) | \Omega \rangle = 0$

$$|\psi\rangle = \sum_n \int d^3p \psi_n(\vec{p}) |p, n\rangle$$

$\psi_n(\vec{p})$: wave packets, $a^\dagger(k) = -i \int d^3x e^{i k x} \overleftrightarrow{\partial}_0 \hat{\psi}(x)$
 $- (k-k_1)^2 / 4\sigma^2$

$$a_1^\dagger \equiv \int d^3k f_1(k) a^\dagger(k) \quad f_1(k) \sim e$$

$\hookrightarrow$ creates a localized particle near $\vec{k}_1$

Thus,

$$\langle \psi | a_1^\dagger(t) | \Omega \rangle = - \sum_n \int d^3p \psi_n^*(p) \int d^3k f_1(k) \int d^3x e^{i k x} \overleftrightarrow{\partial}_0 \langle p, n | \psi(x) | \Omega \rangle$$

Take the asymptotic limits $t \rightarrow \pm \infty$

$$\begin{aligned} \langle \psi | a_1^\dagger(t) | \Omega \rangle &= -i \sum_n \int d^3p \psi_n^*(p) \int d^3k f_1(k) \int d^3x e^{i k x} \overleftrightarrow{\partial}_0 e^{-i p x} A_n(\vec{p}) \\ &= \sum_n \int d^3p \psi_n^*(\vec{p}) \int d^3k f_1(\vec{k}) \int d^3x (p^0 + k^0) e^{i(k-p)x} A_n(\vec{p}) \end{aligned}$$

$$\text{Since } \int d^3x e^{i(\vec{k}-\vec{p})\vec{x}} = (2\pi)^3 \delta^3(\vec{k}-\vec{p})$$

$$p^0 = (\vec{p}^2 + M^2)^{1/2}, \quad M \geq 2m$$

$$k^0 = (\vec{k}^2 + m^2)^{1/2}$$

$$\langle \psi | a_1^\dagger(t) | \Omega \rangle = \sum_n \int d^3p (2\pi)^3 (p^0 + k^0) \psi_n^*(\vec{p}) f_1(\vec{p}) A_n(\vec{p}) e^{i p^0 t}$$

$$p^0 > k^0 \quad \Rightarrow \quad M \geq 2m > m$$

$e^{i(p^0 - k^0)t}$ oscillates rapidly as $t \rightarrow \pm \infty$

By the Riemann-Lebesgue lemma $LHS \rightarrow 0$ $\square$

* LSZ is valid provided $\langle 0 | \phi(x) | 0 \rangle = 0$
 $\langle k | \phi(x) | 0 \rangle = e^{-ikx}$

The Feynmann Propagator:

Free op: $\phi_0(x, t) = \int \widetilde{d^3k} (a_k e^{-ikx} + a_k^\dagger e^{ikx})$

$$\begin{aligned} \langle 0 | \phi_0(x_1) \phi_0(x_2) | 0 \rangle &= \int \widetilde{d^3k_1} \widetilde{d^3k_2} \langle 0 | (a_{k_1} + a_{k_1}^\dagger) (a_{k_2} + a_{k_2}^\dagger) | 0 \rangle \\ &= \int \widetilde{d^3k_1} \widetilde{d^3k_2} \langle 0 | a_{k_1} a_{k_2}^\dagger | 0 \rangle e^{-ik_1 x_1} e^{ik_2 x_2} \\ &= \int \widetilde{d^3k_1} \widetilde{d^3k_2} \langle 0 | a_{k_1} a_{k_2}^\dagger | 0 \rangle^2 e^{ik_2 x_2 - ik_1 x_1} \end{aligned}$$

$$\begin{aligned} \langle 0 | a_{k_1} a_{k_2}^\dagger | 0 \rangle &= (2\pi)^3 \delta^3(\vec{k}_1 - \vec{k}_2) \\ \langle \phi_1, \phi_2 \rangle &= \int \frac{d^3k}{(2\pi)^3} \frac{1}{2\omega_k} e^{ik_1(x_2 - x_1)} \end{aligned}$$

* We are interested in the T.O. product:

$$\Theta(t_i - t_j) = \begin{cases} 1 & , t_i > t_j \\ 0 & , t_i < t_j \end{cases}$$

$\therefore$

$$\begin{aligned} \langle 0 | T \{ \phi_1, \phi_2 \} | 0 \rangle &= \langle 0 | \phi_1 \phi_2 | 0 \rangle \Theta(t_1 - t_2) \\ &\quad + \langle 0 | \phi_2 \phi_1 | 0 \rangle \Theta(t_2 - t_1) \end{aligned}$$

$$\langle 0 | T \{ \phi_1, \phi_2 \} | 0 \rangle = \int \frac{d^3k}{(2\pi)^3} \frac{1}{2\omega_k} \left(e^{ik(x_2 - x_1)} \Theta_{12} + e^{ik(x_1 - x_2)} \Theta_{21} \right)$$

Define $\bar{t} \equiv t_1 - t_2$

$$\langle 0 | T \{ \phi_1, \phi_2 \} | 0 \rangle = \int \frac{d^3k}{(2\pi)^3} \frac{1}{2\omega_k} \left(e^{ik(x_2 - x_1)} \Theta(\bar{t}) + e^{ik(x_1 - x_2)} \Theta(-\bar{t}) \right)$$

Now, take

$k \rightarrow -k$ in the first term

$$= \int \frac{d^3k}{(2\pi)^3} \frac{1}{2\omega_k} e^{-i\vec{k}(\vec{x}_1 - \vec{x}_2)} \left(\underbrace{e^{i\omega_k \tau} \Theta(-\tau)}_{\text{Advanced}} + \underbrace{e^{-i\omega_k \tau} \Theta(\tau)}_{\text{Retarded}} \right)$$

* Now, $\frac{1}{\omega^2 - \omega_k^2 + i\epsilon} = \frac{1}{[\omega - (\omega_k - i\tilde{\epsilon})][\omega - (-\omega_k + i\tilde{\epsilon})]}$

$\mathcal{L}(\epsilon^2) = 0$ a.

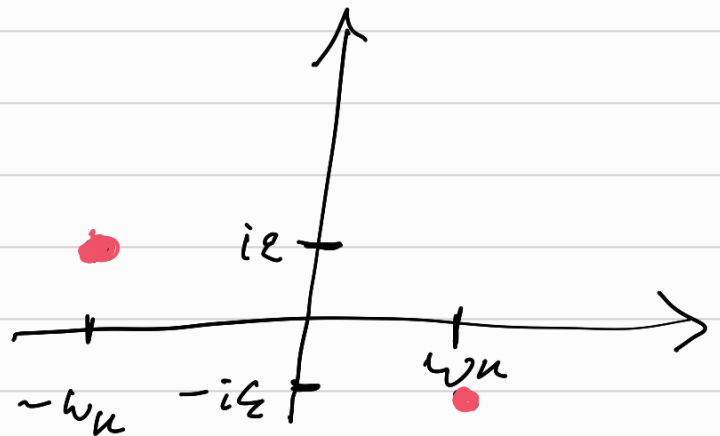
RHS Denom: $\omega^2 + \omega\omega_k - \omega i\epsilon - \omega\omega_k + i\epsilon\omega - \omega_k^2 + i\epsilon\omega_k + i\epsilon\omega_k + \epsilon^2$

$$= \omega^2 - \omega_k^2 + 2i\tilde{\epsilon}\omega_k \leftarrow \equiv \epsilon$$

$$\int_{-\infty}^{\infty} d\omega \frac{e^{i\omega\tau}}{\omega^2 - \omega_k^2 + i\epsilon} = \int_{-\infty}^{\infty} d\omega \frac{e^{i\omega\tau}}{(\quad)(\quad)}$$

ω has poles at

$$\omega = \omega_k + i\epsilon$$

$$\omega = \omega_k - i\epsilon$$


let $\omega = |\omega| e^{i\theta}$

$\tau > 0$: $e^{i\omega\tau} \rightarrow \infty$ as $\omega \rightarrow \infty$

$$e^{(|\omega|\epsilon + i\epsilon)\tau} = e^{i|\omega|\epsilon\tau} \underbrace{e^{-\epsilon\tau}}_{\substack{e^{i(-\omega_k + i\epsilon)\tau} \\ \cdot 2\pi i \Theta(\tau)}}$$

$\therefore \text{LHS} = \int_{-\infty}^{\infty} d\omega \frac{e^{i\omega\tau}}{\omega^2 - \omega_k^2 + i\epsilon} = \int_{-\infty}^{\infty} d\omega \frac{e^{-i\omega_k\tau} e^{i\omega\tau}}{\omega^2 - \omega_k^2 + i\epsilon}$

$$+ (-2\pi i) e^{i\omega_n \tau} \theta(-\tau) e^{i\tau} \frac{1}{2\omega_n}$$

$$\therefore \lim_{\epsilon \rightarrow 0} \int_{-\infty}^{\infty} \frac{d\omega e^{i\omega \tau}}{\omega^2 - \omega_n^2 + i\epsilon} = \frac{-2\pi i}{2\omega_n} \left(e^{-i\omega_n \tau} \theta(-\tau) + e^{i\omega_n \tau} \theta(\tau) \right)$$

Thus

$$\begin{aligned} \langle 0 | T \{ \phi_1 \phi_2 \} | 0 \rangle &= \lim_{\epsilon \rightarrow 0} \int \frac{d^3 k}{(2\pi)^3} \frac{1}{2\omega_n} e^{-i\vec{k} \cdot (\vec{x}_1 - \vec{x}_2)} \\ &\times \int d\omega \frac{-2\pi i \omega_n}{(2\pi i)} \frac{1}{\omega^2 - \omega_n^2 + i\epsilon} e^{-i\omega \tau} \end{aligned}$$

$$\begin{aligned} D_F(x_1, x_2) &= \langle 0 | T \{ \phi_1 \phi_2 \} | 0 \rangle \\ &= \int \frac{d^4 k}{(2\pi)^4} \frac{i}{k^2 - m^2 + i\epsilon} e^{ik(x_1 - x_2)} \end{aligned}$$

(Feynman Propagator)

* $k_0 \neq \sqrt{\vec{k}^2 + m^2}$ as it is a new separate int. variable